Portion Coach PRD

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Overview

Portion Coach tells you how much to eat of the meal in front of you, in under 30 seconds, before you eat it.

Problem. Calorie trackers like Cal AI record what you already ate. Logging is a chore with no payoff in the moment, so most people stop after a couple of weeks.

Insight. In most Pakistani homes you don't choose the menu, the household cook does. The only decision you control is how much you take. No mainstream tracker is built around that decision.

Positioning. Trackers record what you ate. Portion Coach helps you decide what to eat next.

Target user. 18 to 35, lives with family or in a hostel, eats home-cooked desi food, wants to lose or manage weight without cooking separate meals.

Daily trigger. Every meal (2 to 3 times a day). The user opens the app at the table, not afterwards.

Success metrics for the trial

Metric	Target
Time from photo to portion plan	Under 10 seconds
Dish identified correctly on our eval set (50 to 100 desi plates)	80% or more
Items the user had to correct per scan	1 or fewer on average
Full scan flow done without help (new tester)	Under 30 seconds
Testers who log on 5 of 7 days	Tracked, no target yet

Scope

Module 1 is the product and gets most of the 10 days; Modules 2 and 3 are thin layers that prove the roadmap, not full features.

Module	Priority	Share of effort	Done means
1. Portion Coach	Must ship	~60%	Photo to confirmed portion plan, saved to the daily log, tested end to end
2. Duo Streaks	Should ship	~25%	Two real users paired, shared streak updates from real scans
3. Food and Feel	Could ship	~15%	2-tap feel check-in after a meal, one weekly insight from logged data

In scope (v1)

- Email or Google sign-in, a short onboarding (goal, height, weight, activity level)
- Meal photo scan, dish detection, correction chips, portion plan
- Daily calorie budget and today's log
- Pairing with one buddy, shared streak, nudge
- Post-meal feel check-in and a weekly pattern card

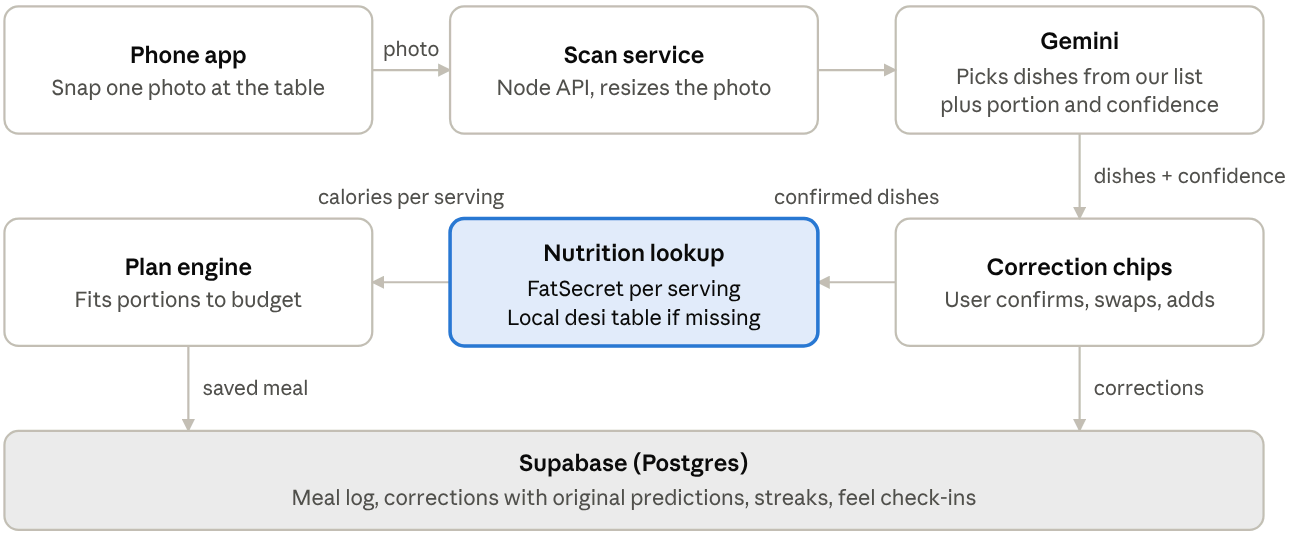
Out of scope (not building)

- Training or fine-tuning our own vision model
- Barcode scanning, packaged food, restaurant menus
- Workouts, step counts, wearables, Apple Health or Google Fit sync
- Groups larger than two, public feeds, leaderboards, chat
- Meal plans, recipes, grocery lists
- Urdu UI (English only for v1; dish names keep their desi names)
- Medical advice, diabetic or allergy modes
- Payments or subscriptions

Guardrail. Every number is labelled as an estimate. The app never gives medical advice.

System overview

Gemini names the dishes; FatSecret or our own table supplies the numbers



scan pipeline · 6 steps into one database

Gemini only returns dish names from our catalog, a portion guess and a confidence score. It never produces calorie numbers.

Stack (proposed)

Layer	Choice	Why
App	React Native (Expo)	Camera-first daily use; Android testable on an emulator
Backend	Node.js + Express, modular monolith	One deployable, one module per feature (scan, nutrition, plan, streaks, feel)
Database, auth, photo storage	Supabase (Postgres)	Provided by 8x; auth and storage included
Hosting	Vercel	Provided by 8x
Vision	Gemini Flash with structured output	No training; answers limited to our dish list

Layer	Choice	Why
Nutrition	FatSecret Platform API + local desi table	Verified data, with a fallback for dishes FatSecret lacks

FatSecret constraints

- The free Basic edition allows 5,000 calls a day, but only the US dataset, English only, and attribution is required.
- Caching is listed only for Premier Free and Premier, so on Basic we should not store their results long term.
- Premier Free is free for students and startups after verification. It removes the call limit and allows caching, still US data only.
- Country datasets such as Pakistan need paid Premier, so dishes like karahi or nihari may be missing or Americanised. The local table covers those.

Recommendation: apply for Premier Free on day 1 and build on Basic until it is approved.

Sources: FatSecret Platform API · FatSecret editions and pricing

Module 1: Portion Coach

The user photographs the food on the table and gets a portion plan that fits what is left of today's budget.

User stories

- As a user, I set my goal once so the app knows my daily calorie budget.
- As a user, I snap the dishes on the table and see what the app detected.
- As a user, I fix a wrong dish or add a missed one in one or two taps.
- As a user, I get a plan in household units (roti, katori, plate) instead of grams.
- As a user, I see how much of today's budget is left after this meal.

Flow

1. Tap Scan, take one photo of the table or plate.
2. Backend sends the photo to Gemini with our dish list; Gemini returns dishes, a rough portion and a confidence score.
3. App shows detected dishes as chips. Low-confidence items ask a question ("Daal or haleem?").

4. User confirms, swaps, removes or adds dishes.
5. Backend gets nutrition per serving for each dish, then builds the plan against the remaining budget.
6. App shows the plan: "1.5 roti, 1 katori karahi, salad as much as you like".
7. User taps "I ate this" (or adjusts the amounts) and the meal is saved to today's log.

Features

- Onboarding: goal (lose, maintain), height, weight, age, activity level. Budget is a standard estimate (Mifflin-St Jeor) with a fixed deficit for "lose".
- Scan screen with camera and gallery upload.
- Correction chips: confirm, swap from the dish list, remove, add.
- Portion units: roti or naan (count), katori (bowl), plate, piece, glass.
- Plan rules: protein and vegetables first, carbs fill the gap, never below a minimum meal size.
- Today screen: budget ring, meals logged, what is left.

Dish catalog

- 30 to 40 common Pakistani dishes, each mapped to a household unit and a grams-per-unit value.
- Each dish stores its nutrition source: FatSecret food id, or our local table if FatSecret has no good match.
- Gemini must answer from this list (structured output with an enum), plus an "unknown" option.

Acceptance criteria

- [ ] Photo to chips in under 10 seconds on a normal connection
- [ ] Unknown or low-confidence dishes always ask, never silently guess
- [ ] Every calorie number on screen shows "estimate"
- [ ] Corrections are saved with the original prediction (future training data)
- [ ] Works when FatSecret is down (falls back to local table)

Edge cases

- No food in the photo: show "No food found, try again".
- Shared dish in the middle of the table: plan covers the user's plate only.
- Budget already used up: show a light plan (salad, raita, half portions), never "don't eat".

Module 2: Duo Streaks

Two friends share one streak, so skipping a day lets your buddy down, not just yourself.

Why it matters. Module 1 gives a reason to open the app at a meal. This module gives a reason to come back on a day the user doesn't feel like it.

User stories

- As a user, I invite one friend with a link or code.
- As a user, I see our shared streak and whether my buddy has logged today.
- As a user, I can nudge my buddy once a day if they haven't logged.

Rules

- A day counts for a person when they save at least one meal through Portion Coach.
- The shared streak grows only when both people count that day.
- Streak resets at midnight Pakistan time (PKT) if either person misses.
- One buddy at a time. Unpairing ends the streak.
- Privacy: the buddy sees "logged today" and the streak, never meals, photos or calories.

Features

- Invite link and 6-character code, accept or decline
- Streak card on the Today screen (count, both avatars, today's status)
- Nudge button (once per day), sent as an in-app notification
- One evening reminder at 9 pm PKT if the user hasn't logged

Acceptance criteria

- [ ] Two real accounts can pair and see the same streak count
- [ ] Streak updates within seconds of a saved meal
- [ ] Reset logic tested across midnight PKT
- [ ] Buddy never sees meal details

Not building: groups, leaderboards, chat, streak freezes, rewards.

Module 3: Food and Feel

A 2-tap check-in after meals that, after a week, tells the user which foods leave them sluggish or energised.

Why it matters. It reuses the meal log Module 1 already collects, so it turns past logs into a payoff and gives a long-term reason to keep logging.

User stories

- As a user, I rate how I feel 1 to 2 hours after a meal in two taps.
- As a user, I see a weekly card like "You felt low energy after 4 of 5 rice-heavy lunches".

Features

- Check-in prompt 90 minutes after a saved meal (in-app; push only if time allows)
- Two questions, one tap each: energy (low, okay, good) and stomach (heavy, fine, light)
- Weekly insight card, built once 7 or more check-ins exist

How the insight works

- Backend groups check-ins by dish category (rice, roti, fried, lentil, meat, sweets).
- A simple rule flags a category when one answer appears in 60% or more of its check-ins, with at least 3 check-ins.
- Gemini only rewrites the flagged pattern into one friendly sentence. It never invents the pattern.
- For the demo, a seeded test account holds 2 weeks of data so the card is visible on day one.

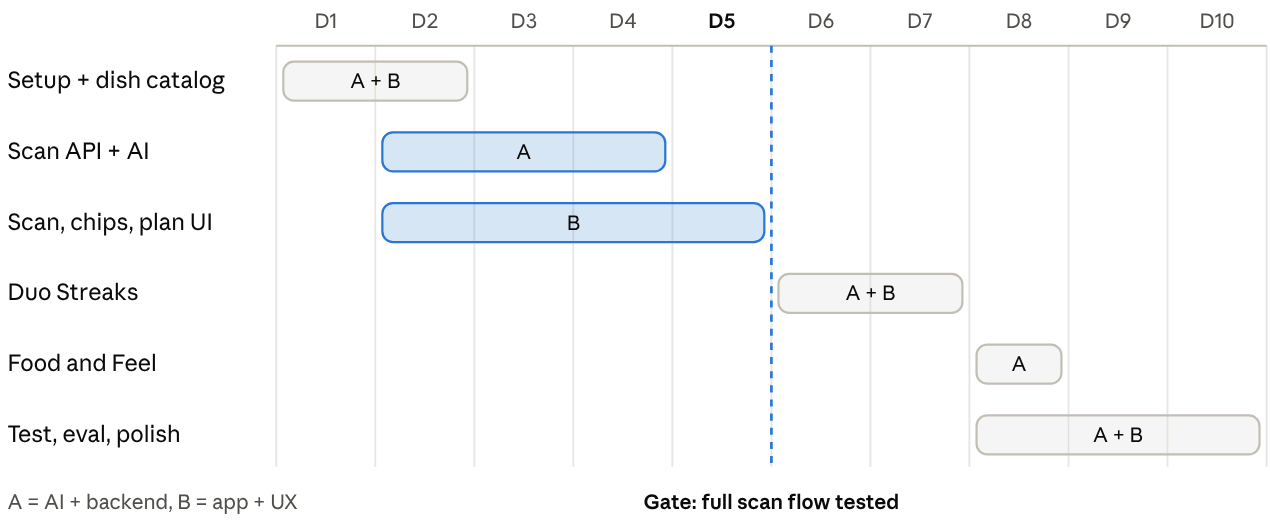
Acceptance criteria

- [ ] Check-in takes 2 taps and under 5 seconds
- [ ] No insight shown with fewer than 3 check-ins in a category
- [ ] Insight text always matches the underlying numbers
- [ ] Copy says "pattern", never "cause" or "diagnosis"

Not building: symptom tracking, sleep or mood journals, medical claims, charts beyond the weekly card.

Plan, risks and open questions

Scan flow works end to end by day 5; days 6 to 10 add layers and polish



10-day plan · 6 workstreams, 1 gate

If the scan flow is not working end to end by day 5, Modules 2 and 3 get cut down before Module 1 does.

Every day, both of us

- Short bullet update in the team Discord channel, in our own words
- Agent logs tracked from day 1 (check the tracking guide first)
- Test every flow we touched ourselves, not just trust the AI

Risks

Risk	Impact	Fallback
FatSecret has no match for a desi dish	Missing or wrong calories	Local desi table, marked in the dish catalog
Gemini confuses look-alike dishes	Wrong plan	Confidence threshold, ask the user, correction chips
Portion guesses are off	Plan feels wrong	Household units the user can adjust in one tap
Basic tier forbids caching	Slow or limited calls	Apply for Premier Free on day 1

Risk	Impact	Fallback
Streak needs two active users	Hard to demo	We two are the first pair; seeded demo pair as backup
Scope creep into Modules 2 and 3	Core feels unfinished	Day 5 gate above

Open questions

- [ ] Mobile app (Expo) or responsive web first? Both of us must be able to test on a phone.
- [ ] Who owns A (AI + backend) and who owns B (app + UX)?
- [ ] Which 30 to 40 dishes go in the catalog?
- [ ] Does 8x provide a template repo, or do we start from scratch?